

DATA MINING

Section 1 — Introduction to Data Mining

1.1 Definition of Data Mining

Data mining is a process that uses statistical, mathematical, and artificial intelligence techniques to extract and identify useful information and patterns from large sets of data.

These patterns can take the form of business rules, affinities, correlations, trends, or prediction models.

Why Is the Term a Misnomer?

The term 'data mining' is considered a misnomer. When gold is mined from rock, we call it 'gold mining' — not 'rock mining.' Similarly, data mining extracts knowledge from data, so it should have been called 'knowledge mining.'

Despite this mismatch, 'data mining' has become the accepted industry term.

Alternative Names for Data Mining

- Knowledge Extraction
- Pattern Analysis
- Data Archaeology
- Information Harvesting
- Pattern Searching
- Data Dredging

1.2 Technical Description

Data mining uses three types of techniques:

- **Statistical techniques** — to find patterns and relationships in data
- **Mathematical techniques** — to model and quantify relationships
- **Artificial Intelligence (AI) techniques** — to learn from data and make predictions

The patterns extracted by data mining include:

- **Business rules** — if-then rules about business behavior
- **Affinities** — items or events that occur together
- **Correlations** — statistical relationships between variables
- **Trends** — patterns that change over time
- **Prediction models** — models that forecast future outcomes

1.3 Characteristics and Objectives of Data Mining

- **Large Databases:** Data is buried deep in very large databases, sometimes spanning years. Data is often cleansed and consolidated into a data warehouse.
- **Architecture:** The environment is usually a client/server or web-based information system architecture.
- **Visualization Tools:** Sophisticated tools, including advanced visualization tools, help uncover buried information in corporate or public records.
- **End-User Empowerment:** End users can ask ad hoc questions and get answers quickly using data drills and query tools — with little or no programming skill.
- **Parallel Processing:** Large search efforts sometimes require parallel processing to handle the volume of data.
- **Software Integration:** Data mining tools integrate readily with spreadsheets and other software development tools.

1.4 Four Major Types of Patterns in Data Mining

1. **Associations** — Find items or events that commonly occur together. Example: In market-basket analysis, beer and diapers are frequently purchased together.
2. **Predictions** — Forecast future outcomes based on past events. Example: Predicting the winner of the Super Bowl, or forecasting the temperature of a specific day.
3. **Clusters** — Identify natural groupings based on characteristics. Example: Assigning customers to different segments based on demographics and past purchase behaviors.
4. **Sequential Relationships** — Discover time-ordered events. Example: A banking customer with a checking account will open a savings account, then an investment account within a year.

Comparison Table — Four Pattern Types:

Pattern Type	Description	Learning Type	Real-World Example
Associations	Items/events that co-occur frequently	Unsupervised	Beer & diapers in market-basket analysis
Predictions	Forecasting future events from past data	Supervised	Predicting Super Bowl winner or weather temperature
Clusters	Natural groupings based on shared characteristics	Unsupervised	Customer segmentation by demographics
Sequential Relationships	Time-ordered events and progressions	Unsupervised	Banking customer: checking → savings → investment account

Section 2 — Data Mining Task Classification

2.1 Three Main Categories of Data Mining Tasks

Prediction

Prediction uses historical data to forecast future outcomes. It answers 'what will happen next?' The output can be a class label (classification) or a numeric value (regression).

Association

Association finds rules about items or events that co-occur. It answers 'what goes together?'

Example: products bought together in one transaction.

Clustering

Clustering groups similar objects without predefined labels. It answers 'what natural groups exist?'

Example: grouping customers by purchasing behavior.

2.2 Supervised vs. Unsupervised Learning

Comparison Table — Supervised vs. Unsupervised Learning:

Feature	Supervised Learning	Unsupervised Learning
Training Data	Contains input variables AND class labels (output)	Contains only input variables — no output labels
Output Variable	Present and known during training	Absent — algorithm discovers structure
Example Methods	Classification, Regression	Clustering, Association
Use Case	Predicting a known outcome (e.g., spam/not spam)	Discovering hidden patterns (e.g., customer segments)
Control Mechanism	Supervised by the class label during learning	No supervising mechanism — uses heuristics

Supervised Learning: The algorithm learns from labeled training data. Both input variables (descriptive attributes) and output variables (class attributes) are provided. The model learns the mapping from input to output.

Unsupervised Learning: The algorithm receives only input variables. It discovers patterns, groupings, or structure on its own — without a supervising output variable.

2.3 Taxonomy of Data Mining Tasks, Methods, and Algorithms

Data mining is organized into a hierarchy of three levels:

- **Tasks** — The high-level goals: Prediction, Association, Clustering
- **Methods** — The approaches used to achieve each task: Classification, Regression, Clustering algorithms, Association rule mining

- **Algorithms** — The specific implementations: Decision Trees, Neural Networks, K-Means, Apriori, etc.

A task can be achieved using multiple methods. Each method can be implemented by multiple algorithms.

Data Mining Tasks & Methods	Data Mining Algorithms	Learning Type
Prediction		
Classification	Decision Trees, Neural Networks, Support Vector Machines, kNN, Naïve Bayes, GA	Supervised
Regression	Linear/Nonlinear Regression, ANN, Regression Trees, SVM, kNN, GA	Supervised
Time series	Autoregressive Methods, Averaging Methods, Exponential Smoothing, ARIMA	Supervised
Association		
Market-basket	Apriori, OneR, ZeroR, Eclat, GA	Unsupervised
Link analysis	Expectation Maximization, Apriori Algorithm, Graph-Based Matching	Unsupervised
Sequence analysis	Apriori Algorithm, FP-Growth, Graph-Based Matching	Unsupervised
Segmentation		
Clustering	k-means, Expectation Maximization (EM)	Unsupervised
Outlier analysis	k-means, Expectation Maximization (EM)	Unsupervised

Section 3 — Data Mining Applications

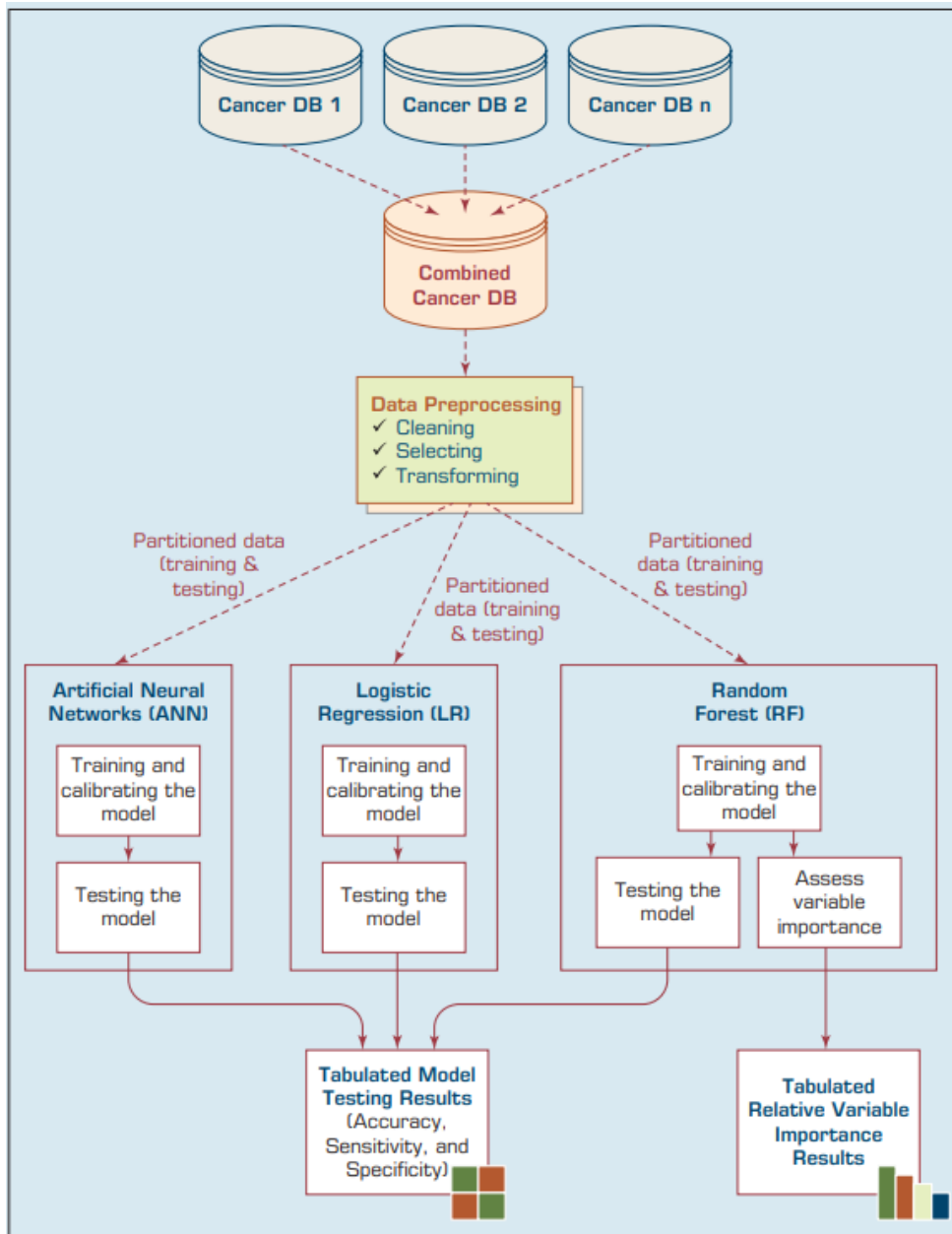
3.1 Industry-Wise Applications

Data mining is applied across virtually every industry. The table below summarizes major application areas and their use cases.

Industry / Domain	Data Mining Application
Customer Relationship Management (CRM)	Identify valuable customers, predict churn, personalize marketing campaigns
Banking	Credit scoring, fraud detection, loan default prediction, customer segmentation
Retailing and Logistics	Market-basket analysis, inventory management, demand forecasting, store layout optimization
Manufacturing and Production	Quality control, predictive maintenance, defect detection, production scheduling
Brokerage and Securities Trading	Stock price prediction, algorithmic trading, portfolio risk analysis
Insurance	Fraud detection, risk assessment, policy pricing, claims processing optimization
Computer Hardware and Software	Fault prediction, usage pattern analysis, software bug detection
Government and Defense	Crime pattern analysis, national security threat detection, resource allocation
Travel Industry (Airlines, Hotels, Rental Cars)	Yield management, customer loyalty analysis, demand prediction, pricing optimization
Healthcare	Patient risk stratification, hospital readmission prediction, resource planning
Medicine	Disease diagnosis, drug discovery, clinical trial analysis, comorbidity investigation
Entertainment Industry	Content recommendation (Netflix, Spotify), audience preference analysis
Sports	Player performance analysis, game strategy optimization, injury prediction

Section 4 — The Data Mining Process (CRISP-DM)

CRISP-DM stands for **Cross-Industry Standard Process for Data Mining**. It is the most widely used methodology for data mining projects. It provides a structured, repeatable process with six steps.



Step 1: Business Understanding

Definition: The process of understanding the business need and defining the objectives of the data mining study.

Core Concept: This is the most critical step. All subsequent steps depend on a clear business objective. Without it, the project has no direction.

Key activities:

- Identify the managerial need for new knowledge
- Write an explicit specification of the business objective
- Ask: What problem are we solving? What decision does this support?

Example business questions:

- *"What are the common characteristics of customers we have lost to our competitors recently?"*
- *"What are typical profiles of our customers, and how much value does each provide?"*

Exam Note: This step defines the entire direction of the project. Always covered in exams.

Step 2: Data Understanding

Definition: The process of identifying, collecting, and exploring the data relevant to the business objective.

Core Concept: Different business tasks require different data. The analyst must understand both the data sources and the variables.

Key considerations:

- **Location:** Where is the data stored and in what form?
- **Collection process:** Is data collection automated or manual?
- **Collectors:** Who collects the data and how often is it updated?
- **Synonymous variables:** Are there variables with the same meaning but different names?
- **Homonymous variables:** Are there variables with the same name but different meanings?
- **Variable independence:** Do variables overlap or conflict with each other?

Example: A retail project may use demographics, credit card transactions, and socioeconomic attributes to identify spending behaviors of female shoppers who purchase seasonal clothing.

Step 3: Data Preparation (Data Preprocessing)

Definition: The process of transforming raw data into a clean, analysis-ready dataset.

Core Concept: Real-world data is rarely perfect. It must be cleaned, transformed, and structured before analysis.

Time Commitment: Data preparation consumes approximately **80%** of total project time.

Three key problems with real-world data:

Data Quality Problem	Definition	Example
Incomplete Data	Missing attribute values, missing attributes, or only aggregate data is available	A customer record with no age or income field
Noisy Data	Data containing errors, outliers, or incorrect values	A transaction record showing a purchase amount of -\$500

Data Quality Problem	Definition	Example
Inconsistent Data	Discrepancies in codes, names, or formats across records	One table uses 'M/F', another uses 'Male/Female' for gender

Exam Note: *Data Preparation is the most time-consuming step in CRISP-DM — always appears in exams.*

Step 4: Model Building

Definition: The process of selecting and applying data mining techniques to the prepared dataset.

Core Concept: No single algorithm is universally best for all problems. Multiple models must be built and compared.

Key activities:

- Select candidate modeling techniques appropriate for the business task
- Apply multiple techniques and compare their results
- Use a well-defined experimentation and assessment strategy
- Identify the best-performing model for the given purpose

Exam Note: *Model selection depends on the business objective — there is no single best algorithm.*

Step 5: Testing and Evaluation

Definition: The process of assessing the developed models for accuracy, quality, and relevance to business objectives.

Core Concept: A model must be evaluated against both technical metrics and business requirements.

Key activities:

- Assess whether the model meets the business objectives
- Measure the degree to which it succeeds
- Decide if additional models need to be developed
- Test models in real-world scenarios when time and budget allow

Step 6: Deployment

Definition: The process of organizing and presenting the discovered knowledge so end users can understand and use it.

Core Concept: Data mining does not end at model building. The knowledge gained must be delivered to the people who need it.

Key activities:

- Organize and present knowledge in a user-understandable format
- Integrate the model into existing business processes or systems

- Maintain the deployed model — because business data changes constantly
- Update models when the underlying data patterns shift

Exam Note: *Deployment includes both presentation of knowledge AND ongoing maintenance of models.*

CRISP-DM Summary Table:

Step No.	Step Name	Key Activity	Primary Output
1	Business Understanding	Define business objectives and project goals	Project plan and objective statement
2	Data Understanding	Identify, collect, and explore relevant data	Data inventory and exploration report
3	Data Preparation	Clean, transform, and prepare data (80% of time)	Clean, analysis-ready dataset
4	Model Building	Select and apply modeling techniques	Multiple trained candidate models
5	Testing and Evaluation	Assess model accuracy and business fit	Evaluated and selected model
6	Deployment	Present and maintain knowledge for end users	Deployed solution with maintenance plan

Section 5 — Other Data Mining Processes and Methodologies

5.1 SEMMA Process

SEMMA was developed by **SAS Institute**. It is a data mining process methodology tied to the SAS software environment.

SEMMA stands for:

Letter	Phase	Description
S	Sample	Select a representative sample of the data to work with
E	Explore	Visualize and explore the data to identify patterns and anomalies
M	Modify	Create, select, and transform variables for modeling
M	Model	Apply modeling techniques to the modified data
A	Assess	Evaluate the usefulness and reliability of the model

Exam Note: *SEMMA is tool-specific (SAS), while CRISP-DM is tool-independent and industry-neutral.*

5.2 KDD — Knowledge Discovery in Databases

KDD stands for **Knowledge Discovery in Databases**.

Definition (Fayyad et al., 1996): KDD is the process of using data mining methods to find useful information and patterns in data. It is a comprehensive process that encompasses data mining as one of its steps.

KDD vs. Data Mining: Key Distinction

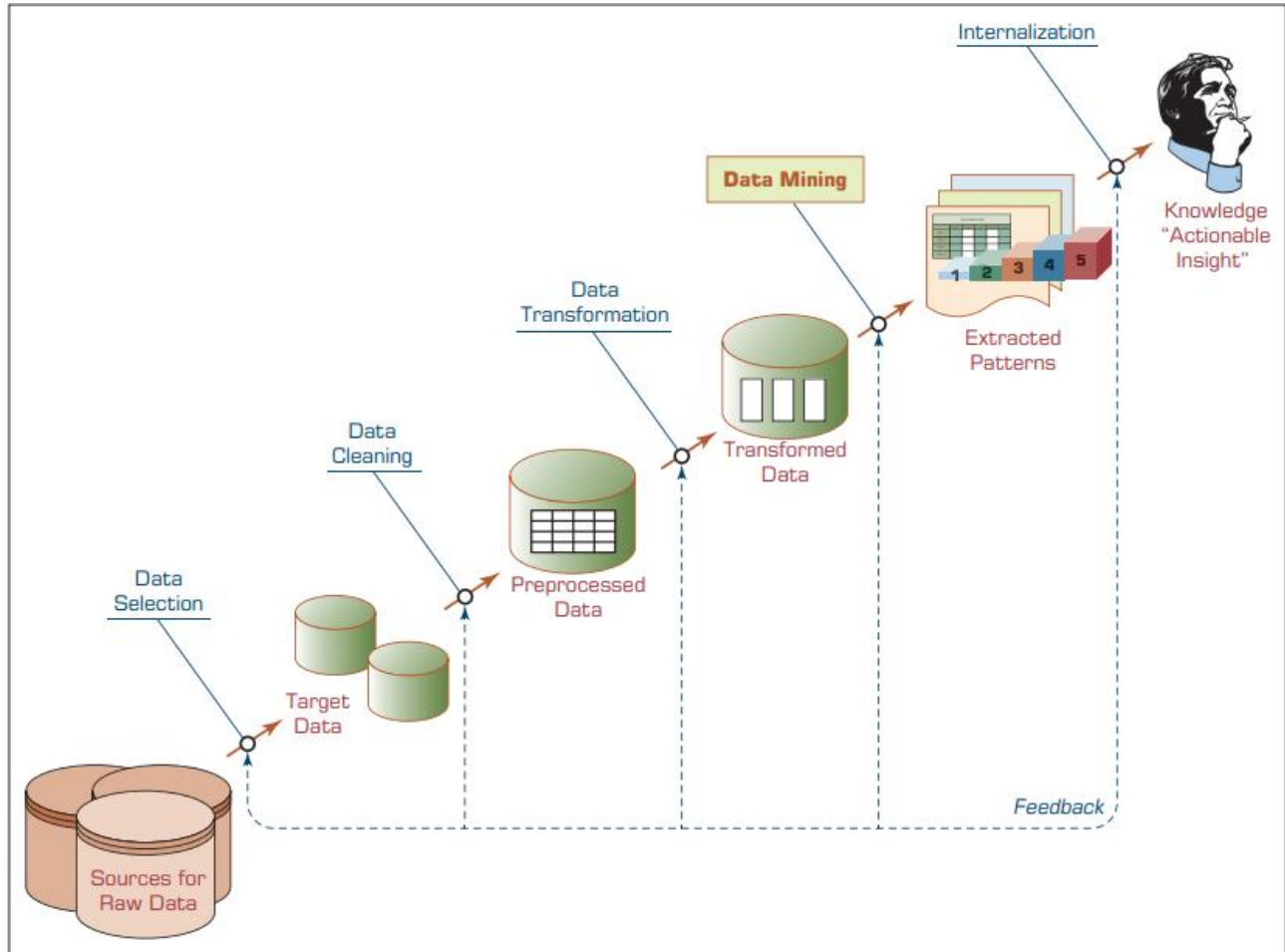
- **KDD:** A comprehensive, end-to-end process for discovering knowledge from organizational data
- **Data Mining:** The specific algorithmic step within KDD where patterns are extracted from data

In other words, data mining is one step inside the larger KDD process.

Role of the Enterprise Data Warehouse: The enterprise data warehouse enables KDD to be implemented efficiently. It provides a single, integrated source of organizational data to be mined.

Five Steps of KDD (Dunham, 2003)

5. Data Selection — Choose the relevant subset of data
6. Data Preprocessing — Clean and prepare the selected data
7. Data Transformation — Convert data into a suitable format for mining
8. Data Mining — Apply algorithms to extract patterns
9. Interpretation / Evaluation — Interpret patterns and evaluate usefulness



Comparison Table — CRISP-DM vs. SEMMA vs. KDD:

Feature	CRISP-DM	SEMMA	KDD
Number of Steps	6	5	5
Developed By	Consortium (DaimlerChrysler, SPSS, NCR, etc.)	SAS Institute	Fayyad et al. (1996)
Tool Dependency	Tool-independent	SAS-specific	Tool-independent
Scope	Full project lifecycle from business need to deployment	Data-centric — focuses on modeling process	Full knowledge discovery process
Primary Focus	Business understanding and deployment	Modeling and assessment	Knowledge extraction from databases

Section 6 — Data Mining Methods

Four main data mining methods are used in practice. Most software tools implement more than one method.

6.1 Classification

Definition: Classification is a supervised learning method that learns patterns from labeled historical data and assigns new instances to predefined classes.

Core Concept: The algorithm learns from past data where the class label is known. It then uses that knowledge to predict the class of new, unlabeled data.

How It Works

10. Collect training data with known class labels
11. Train the classification model on the training data
12. Evaluate the model on a holdout test set
13. Deploy the model to classify new, unknown instances

Examples of Classification Tasks

- Credit approval — Good credit risk / Bad credit risk
- Weather prediction — Sunny / Rainy / Cloudy
- Fraud detection — Yes / No
- Store location selection — Good / Moderate / Bad
- Customer churn — Likely to leave / Not likely to leave

Classification vs. Regression — Critical Distinction

Classification: Output is a **class label** (a category). Example: predicting weather as 'sunny', 'rainy', or 'cloudy'.

Regression: Output is a **numeric value**. Example: predicting the exact temperature as 68°F.

Exam Note: If output = class label → Classification. If output = number → Regression. This distinction is frequently tested.

6.2 Regression

Definition: Regression is a supervised learning method that predicts a continuous numeric output value.

Core Concept: Regression models the relationship between input variables and a continuous output. It is used when the prediction target is a number, not a category.

Example: Predicting the temperature of a specific day (e.g., 68°F) based on historical weather data.

6.3 Clustering

Definition: Clustering is an unsupervised learning method that discovers natural groupings of objects based on their similarity.

Core Concept: No class labels are provided. The algorithm uses heuristics — such as multidimensional distance measures — to group similar objects together.

Example: Assigning customers to different segments based on their demographics and past purchase behaviors.

Key Difference: Clustering vs. Classification

- **Classification:** Supervised — the class labels are known during training
- **Clustering:** Unsupervised — no class labels; groups are discovered automatically
- **Classification:** Has a controlling mechanism (the class label)
- **Clustering:** No controlling mechanism — uses heuristics like distance measures

6.4 Association

Definition: Association is a data mining method that finds rules about items or events that frequently co-occur.

Core Concept: Association rule mining identifies if-then relationships between items. It is the method behind market-basket analysis.

Example: Customers who buy beer also frequently buy diapers. Rule: 'beer → diapers' with high confidence.

Comprehensive Comparison Table — Four Data Mining Methods:

Feature	Classification	Regression	Clustering	Association
Learning Type	Supervised	Supervised	Unsupervised	Unsupervised
Output Type	Class label (category)	Numeric value	Group/cluster membership	If-then rules
Class Labels in Training	Required	Required (numeric)	Not required	Not required
Algorithm Type	Decision Trees, Neural Nets, SVM	Linear/Logistic Regression	K-Means, DBSCAN	Apriori, FP-Growth
Real-World Example	Fraud detection (yes/no)	Temperature prediction (68°F)	Customer segmentation	Market-basket analysis

Section 7 — Classification in Depth: Model Assessment

7.1 The Two-Step Classification Methodology

Every classification process follows two main phases:

Phase 1: Model Development (Training)

- A collection of training data is used — including both input variables and known class labels
- The algorithm learns the mapping from input variables to class labels
- The output of this phase is a trained classification model

Phase 2: Model Testing and Deployment

- The trained model is tested against a holdout sample (test dataset not used in training)
- Accuracy is assessed by comparing predicted labels vs. actual labels
- If accuracy is satisfactory, the model is deployed to classify new, unknown instances

7.2 Five Factors for Assessing Classification Models

When evaluating a classification model, five key factors are considered:

1. Predictive Accuracy

Definition: The model's ability to correctly predict the class label of new or previously unseen data. Predictive accuracy is the most commonly used assessment factor.

How to compute:

- Take a test dataset with known class labels
- Run the model to predict class labels for each record
- Compare predicted labels to actual labels

Accuracy Rate = (Correctly Classified Instances / Total Instances) × 100%

Variable definitions:

- **Correctly Classified Instances:** Number of test records where the model's prediction matched the actual class label
- **Total Instances:** Total number of records in the test dataset

2. Speed

Definition: The computational cost involved in generating and using the model.

Faster models are preferred. Speed matters both during training and during prediction.

3. Robustness

Definition: The model's ability to make reasonably accurate predictions even when given noisy data, data with missing values, or data with errors.

A robust model maintains performance despite imperfect input data.

4. Scalability

Definition: The ability to construct a prediction model efficiently when given very large amounts of data.

A scalable model can handle growing datasets without a significant drop in performance or unacceptable increases in computation time.

5. Interpretability

Definition: The level of understanding and insight provided by the model — how and why the model makes certain predictions.

High interpretability means stakeholders can understand and trust the model's decisions.

Summary Table — Model Assessment Factors:

Factor	Definition	Why It Matters
Predictive Accuracy	% of correct predictions on unseen data	Primary measure of model quality
Speed	Computational cost of training and prediction	Affects real-time usability in production
Robustness	Accuracy maintained despite noisy/missing data	Real-world data is always imperfect
Scalability	Efficiency with very large datasets	Enterprise data can be massive
Interpretability	Understandability of model decisions	Builds stakeholder trust and regulatory compliance

7.3 The Confusion Matrix

Definition: A confusion matrix (also called a **classification matrix** or **contingency table**) is a table that summarizes the performance of a classification model.

It is the primary source for accuracy estimation in classification problems.

Structure for a Two-Class Problem

The rows represent the actual class. The columns represent the predicted class.

	Predicted: POSITIVE	Predicted: NEGATIVE
Actual: POSITIVE	True Positive (TP)	False Negative (FN)
Actual: NEGATIVE	False Positive (FP)	True Negative (TN)

Explanation of Each Cell

- **True Positive (TP):** The model predicted POSITIVE and the actual label is POSITIVE. Correct prediction.
- **True Negative (TN):** The model predicted NEGATIVE and the actual label is NEGATIVE. Correct prediction.
- **False Positive (FP):** The model predicted POSITIVE but the actual label is NEGATIVE. Incorrect — also called Type I Error.
- **False Negative (FN):** The model predicted NEGATIVE but the actual label is POSITIVE. Incorrect — also called Type II Error.

Key Formulas Derived from the Confusion Matrix

Metric	Formula	Intuition	Practical Meaning
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall correct predictions / all predictions	What % of all predictions were correct?
Precision	$TP / (TP + FP)$	Correct positive predictions / all predicted positives	When the model says positive, how often is it right?
Recall (Sensitivity)	$TP / (TP + FN)$	Correct positive predictions / all actual positives	Of all actual positives, how many did we catch?
Specificity	$TN / (TN + FP)$	Correct negative predictions / all actual negatives	Of all actual negatives, how many did we correctly reject?
F1 Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Harmonic mean of Precision and Recall	Balances precision and recall — useful for imbalanced datasets

		True/Observed Class	
		Positive	Negative
Predicted Class	Positive	True Positive Count (TP)	False Positive Count (FP)
	Negative	False Negative Count (FN)	True Negative Count (TN)

Section 8 — Exam Preparation

8.1 Quick Revision Notes

Section 1 — Introduction

- Data mining extracts knowledge from large datasets using statistical, mathematical, and AI techniques.
- 'Data mining' is a misnomer — knowledge mining would be more accurate.
- Alternative names: knowledge extraction, pattern analysis, data archaeology, information harvesting.
- Four pattern types: Associations, Predictions, Clusters, Sequential Relationships.

Section 2 — Task Classification

- Three main tasks: Prediction, Association, Clustering.
- Supervised learning uses labeled data (input + output). Unsupervised uses only input.
- Classification and regression are supervised; clustering and association are unsupervised.

Section 3 — Applications

- Data mining is applied in 13+ industries including banking, healthcare, retail, sports, and government.

Section 4 — CRISP-DM

- CRISP-DM has 6 steps: Business Understanding, Data Understanding, Data Preparation, Model Building, Testing & Evaluation, Deployment.
- Step 3 (Data Preparation) consumes ~80% of total project time.
- Data problems: Incomplete, Noisy, Inconsistent.

Section 5 — SEMMA and KDD

- SEMMA: Sample → Explore → Modify → Model → Assess. Developed by SAS Institute. Tool-specific.
- KDD: Selection → Preprocessing → Transformation → Mining → Interpretation. Tool-independent.
- Data mining is ONE step inside the broader KDD process.

Section 6 — Methods

- Classification = supervised, outputs class label.
- Regression = supervised, outputs numeric value.
- Clustering = unsupervised, discovers natural groups.
- Association = unsupervised, finds co-occurring items.

Section 7 — Model Assessment

- Five assessment factors: Predictive Accuracy, Speed, Robustness, Scalability, Interpretability.
- Confusion matrix shows TP, TN, FP, FN for a two-class problem.

- Key formulas: Accuracy, Precision, Recall, Specificity, F1 Score.

8.2 Important Definitions (Glossary)

Term	Definition
Data Mining	A process using statistical, mathematical, and AI techniques to extract useful patterns and knowledge from large datasets
Knowledge Discovery in Databases (KDD)	A comprehensive end-to-end process for discovering useful knowledge from organizational data; data mining is one step within it
CRISP-DM	Cross-Industry Standard Process for Data Mining; a 6-step industry-standard methodology for executing data mining projects
SEMMA	Sample, Explore, Modify, Model, Assess; a 5-step data mining process developed by SAS Institute
Classification	A supervised data mining method that learns from labeled data to assign class labels to new instances
Regression	A supervised method that predicts a continuous numeric output variable
Clustering	An unsupervised method that discovers natural groupings in data without predefined class labels
Association	An unsupervised method that finds rules about items or events that frequently co-occur
Supervised Learning	Learning from labeled training data that includes both input variables and the output class label
Unsupervised Learning	Learning from data that includes only input variables; the algorithm discovers structure independently
Training Data	The labeled dataset used to train a classification or regression model
Holdout Sample	A test dataset not used in training, reserved for evaluating model accuracy
Confusion Matrix	A table summarizing classification performance showing TP, TN, FP, FN counts
True Positive (TP)	A prediction where the model correctly predicted the positive class
False Positive (FP)	A prediction where the model incorrectly predicted positive when the actual label was negative (Type I Error)
Predictive Accuracy	The percentage of test instances correctly classified by the model
Robustness	The model's ability to maintain accuracy despite noisy, missing, or erroneous input data
Scalability	The model's ability to perform efficiently when applied to very large datasets
Interpretability	The degree to which the model's decision process can be understood by humans

Term	Definition
Data Preprocessing	The transformation of raw data into a clean, analysis-ready dataset (Step 3 of CRISP-DM)
Data Warehouse	A centralized integrated repository of organizational data used as the input for data mining and KDD
Market-Basket Analysis	A data mining application that finds associations between products purchased together in the same transaction
Pattern	A recurring regularity or rule discovered in data by a data mining algorithm
Prediction	Forecasting the value of a variable (class label or numeric) based on historical patterns
Heuristic	A problem-solving approach that uses practical rules or distance measures to find approximate solutions, used in clustering

8.3 Important Formulas

Accuracy Rate (Basic)

$$\text{Accuracy Rate} = (\text{Correctly Classified Instances} / \text{Total Instances}) \times 100\%$$

Variables: Correctly Classified Instances = number of test records where prediction matched actual label; Total Instances = total test records.

Intuition: What percentage of all predictions were correct?

Confusion Matrix Formulas

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

All correct predictions divided by all predictions. Best general performance measure.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Of all predicted positives, how many were actually positive? High precision = few false alarms.

$$\text{Recall (Sensitivity)} = \text{TP} / (\text{TP} + \text{FN})$$

Of all actual positives, how many did the model catch? High recall = few missed positives.

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

Of all actual negatives, how many did the model correctly reject? High specificity = few false positives.

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Harmonic mean of Precision and Recall. Useful when the dataset is imbalanced (one class is much more frequent than the other).

8.4 Important Comparisons

Data Mining vs. KDD

Feature	Data Mining	KDD
Definition	Algorithmic step that extracts patterns from data	Comprehensive process for knowledge discovery
Scope	Narrow — one step in a process	Broad — encompasses data mining
Steps	Applies algorithms to find patterns	Selection, Preprocessing, Transformation, Mining, Interpretation
Focus	Pattern extraction	End-to-end knowledge discovery

Classification vs. Regression

Feature	Classification	Regression
Output Type	Class label (category)	Numeric value
Learning Type	Supervised	Supervised
Example	Weather: sunny/rainy/cloudy	Temperature: 68°F
Common Algorithms	Decision Trees, Neural Nets, SVM	Linear Regression, Logistic Regression

Classification vs. Clustering

Feature	Classification	Clustering
Learning Type	Supervised	Unsupervised
Class Labels in Training	Required	Not required
Control Mechanism	Class label supervises learning	Heuristics (e.g., distance measures)
Output	Predicts known class labels	Discovers new groups/clusters
Example	Spam detection (spam/not spam)	Customer segmentation

Supervised vs. Unsupervised Learning

Feature	Supervised Learning	Unsupervised Learning
Training Data	Input + Output (labeled)	Input only (unlabeled)
Output Variable	Present and known	Absent
Example Methods	Classification, Regression	Clustering, Association
Goal	Predict a known output	Discover hidden structure

Feature	Supervised Learning	Unsupervised Learning
Control	Supervised by class labels	No external supervision

CRISP-DM vs. SEMMA vs. KDD

Feature	CRISP-DM	SEMMA	KDD
Steps	6	5	5
Developed By	Industry consortium	SAS Institute	Fayyad et al. (1996)
Tool Dependency	Independent	SAS-specific	Independent
Business Orientation	Strong (Step 1 = Business Understanding)	Weak	Moderate
Includes Deployment	Yes	No	No (ends at interpretation)

8.5 Frequently Asked Exam Questions

Q1. Q: What is data mining? Why is it considered a misnomer?

A: Data mining is a process that uses statistical, mathematical, and AI techniques to extract useful patterns and knowledge from large datasets. It is called a misnomer because, just as gold mining extracts gold (not rock), data mining extracts knowledge (not data). It should more accurately be called 'knowledge mining.'

Q2. Q: What are the four major types of patterns identified by data mining?

A: The four types are: (1) Associations — items that co-occur (e.g., beer and diapers); (2) Predictions — forecasting future events from past data; (3) Clusters — natural groupings of similar objects; (4) Sequential Relationships — time-ordered event progressions.

Q3. Q: What is the most time-consuming step in CRISP-DM and why?

A: Step 3: Data Preparation. It consumes approximately 80% of the total project time because real-world data is incomplete (missing values), noisy (errors, outliers), and inconsistent (discrepancies in codes or names).

Q4. Q: What is the difference between classification and regression?

A: Both are supervised learning methods. Classification predicts a class label (a category), such as 'sunny' or 'rainy.' Regression predicts a numeric value, such as a temperature of 68°F.

Q5. Q: What is the difference between classification and clustering?

A: Classification is supervised — it uses labeled training data with known class labels to predict labels for new data. Clustering is unsupervised — it has no class labels and discovers natural groupings on its own using heuristics like distance measures.

Q6. Q: What is a confusion matrix and what does it measure?

A: A confusion matrix is a table that summarizes the performance of a classification model. It shows the counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). It is the primary source for accuracy estimation in classification.

Q7. Q: What does SEMMA stand for?

A: S — Sample, E — Explore, M — Modify, M — Model, A — Assess. SEMMA was developed by SAS Institute.

Q8. Q: What is KDD and how does it differ from data mining?

A: KDD (Knowledge Discovery in Databases) is a comprehensive end-to-end process for discovering knowledge from data. Data mining is just one step within KDD — the algorithmic step where patterns are extracted. KDD also includes data selection, preprocessing, transformation, and interpretation.

Q9. Q: What are the five factors for assessing a classification model?

A: The five factors are: (1) Predictive Accuracy, (2) Speed, (3) Robustness, (4) Scalability, and (5) Interpretability.

Q10. Q: What formula is used to compute accuracy rate?

A: Accuracy Rate = (Correctly Classified Instances / Total Instances) × 100%. Correctly Classified Instances is the count of test records where the predicted label matched the actual label. Total Instances is the total number of test records.

Q11. Q: What is supervised learning?

A: Supervised learning is a machine learning approach where the training data includes both the input variables (descriptive attributes) and the output variable (class label). The algorithm learns the mapping from input to output using these labeled examples.

Q12. Q: What is market-basket analysis?

A: Market-basket analysis is a data mining application that finds associations between products purchased together in the same transaction. A classic example is the association between beer and diapers being purchased together.

Q13. Q: What is the role of the enterprise data warehouse in KDD?

A: The enterprise data warehouse serves as a single, integrated source of organizational data. It enables KDD to be implemented efficiently because all relevant data can be accessed from one central location rather than scattered across multiple systems.

Q14. Q: What is parallel processing and when is it used in data mining?

A: Parallel processing uses multiple processors simultaneously to perform computations. In data mining, it is used when datasets are very large and the search effort is massive — where a single processor would be too slow to complete the analysis in a reasonable time.

Q15. Q: What is the deployment step in CRISP-DM?

A: Deployment is the final step in CRISP-DM. It involves organizing and presenting the discovered knowledge so end users can understand and benefit from it. It also includes maintaining the deployed models, as business data constantly changes and models may need to be updated.

8.6 Viva Questions with Answers

V1. Q: Explain the CRISP-DM process with a real-world example.

Detailed Answer: CRISP-DM has 6 steps. Using a bank fraud detection project as an example: (1) Business Understanding — the bank wants to detect fraudulent transactions; (2) Data Understanding — collect transaction history, customer data, and account activity; (3) Data Preparation — clean missing values and encode categorical variables; (4) Model Building — train a classification model (e.g., Decision Tree); (5) Testing and Evaluation — test accuracy using a holdout set and confusion matrix; (6) Deployment — integrate the model into the bank's transaction processing system and monitor it for performance degradation.

V2. Q: Why does data preprocessing consume 80% of a data mining project's time?

Detailed Answer: Real-world data is almost never clean or ready for analysis. Three fundamental problems must be solved: (1) Incomplete data — records with missing fields must be handled (imputed or removed); (2) Noisy data — outliers and errors must be detected and corrected; (3) Inconsistent data — naming conventions, codes, and formats must be standardized. Each problem requires careful, labor-intensive investigation and manual or semi-automated correction across potentially millions of records.

V3. Q: How do you evaluate a classification model?

Detailed Answer: Use the confusion matrix to compute: Accuracy (overall correctness), Precision (correctness among positive predictions), Recall (coverage of actual positives), and F1 Score (balance of Precision and Recall). Also evaluate Speed (how fast does it predict?), Robustness (does it handle noisy data?), Scalability (can it handle large datasets?), and Interpretability (can users understand why it made its decisions?).

V4. Q: What is the difference between supervised and unsupervised learning?

Detailed Answer: Supervised learning uses labeled training data — both input variables and the output class label are provided. The algorithm learns a function from input to output. Unsupervised learning uses only input variables — no labels are provided. The algorithm discovers patterns or groupings on its own using heuristics. Classification and regression are supervised; clustering and association are unsupervised.

V5. Q: Explain the confusion matrix with an example.

Detailed Answer: Suppose we build a model to detect email spam (Positive = Spam, Negative = Not Spam). We test on 100 emails: 40 are spam, 60 are not. The model predicts: TP=35 (correctly identified spam), TN=55 (correctly identified non-spam), FP=5 (marked as spam but weren't), FN=5 (missed spam). Accuracy = $(35+55)/100 = 90\%$. Precision = $35/40 = 87.5\%$. Recall = $35/40 = 87.5\%$. These numbers tell us how well the model performs for each type of error.

V6. Q: What are associations in data mining? Give an example.

Detailed Answer: Associations identify items or events that frequently occur together. The classic example is market-basket analysis: a supermarket analyzes transaction data and discovers that

customers who buy beer also frequently buy diapers. This association rule (beer → diapers) can be used to optimize store layout or promotional offers.

V7. Q: What is the difference between KDD and CRISP-DM?

Detailed Answer: KDD is a conceptual framework that describes the process of knowledge discovery from databases — it is academic and theoretical. CRISP-DM is a practical, industry-standard methodology for executing data mining projects — it is operational and includes business-oriented steps like Business Understanding and Deployment. CRISP-DM has 6 steps; KDD has 5. Both are tool-independent.

V8. Q: Why is Business Understanding considered the most important step in CRISP-DM?

Detailed Answer: Without a clear business objective, all subsequent steps are misdirected. The data collected, the preprocessing applied, the model built, and the deployment strategy all depend on what problem the project is solving. A model that is technically excellent but answers the wrong business question has zero value. Business Understanding anchors the entire project.

V9. Q: What is the taxonomy of data mining? Explain its three levels.

Detailed Answer: The taxonomy organizes data mining into three levels: (1) Tasks — the high-level goals (Prediction, Association, Clustering); (2) Methods — the approaches for achieving each task (Classification, Regression, K-Means, Apriori, etc.); (3) Algorithms — specific implementations of each method (Decision Trees, Neural Networks, SVM, K-Means, Apriori). One task can have multiple methods; one method can be implemented by multiple algorithms.

V10. Q: What makes a data mining model robust and why does it matter?

Detailed Answer: A robust model maintains reasonable accuracy even when given noisy data, missing values, or erroneous input. This matters because real-world production data is always imperfect. A model that performs well only on clean test data but fails in production due to data quality issues is not usable. Robustness is tested by deliberately introducing noise into the test dataset and measuring accuracy degradation.

8.7 One-Page Revision Summary

Data Mining: A process using statistical, mathematical, and AI techniques to extract useful patterns and knowledge from large datasets.

4 Pattern Types

- **Associations:** Items that co-occur (e.g., beer & diapers)
- **Predictions:** Forecasting future outcomes from past data
- **Clusters:** Natural groupings based on similarity
- **Sequential Relationships:** Time-ordered event chains

3 Data Mining Task Categories

- Prediction | Association | Clustering

6 CRISP-DM Steps

14. **Business Understanding:** Define project goals and business objectives
15. **Data Understanding:** Identify, collect, and explore relevant data
16. **Data Preparation:** Clean and transform data (80% of time)
17. **Model Building:** Select and apply modeling techniques
18. **Testing & Evaluation:** Assess accuracy and business fit
19. **Deployment:** Present knowledge and maintain models

4 Data Mining Methods

- **Classification:** Supervised → outputs class label
- **Regression:** Supervised → outputs numeric value
- **Clustering:** Unsupervised → discovers natural groups
- **Association:** Unsupervised → finds co-occurring items

5 Model Assessment Factors

- **Accuracy:** % of correct predictions
- **Speed:** Computational efficiency
- **Robustness:** Handles noisy/missing data
- **Scalability:** Works with large datasets
- **Interpretability:** Model decisions are understandable

Confusion Matrix Formulas

$$\begin{aligned} \text{Accuracy} &= (TP+TN) / (TP+TN+FP+FN) & \text{Precision} &= TP / (TP+FP) \\ \text{Recall} &= TP / (TP+FN) & \text{F1} &= 2 \times (P \times R) / (P+R) \end{aligned}$$

Key Comparisons

- **Classification vs. Clustering:** Classification is supervised (has labels); Clustering is unsupervised (discovers groups)
- **Supervised vs. Unsupervised:** Supervised has output labels in training data; Unsupervised has only input data